



AI 2026

IA-II

Ethical & Explainable AI

Why Accuracy Is Not Enough!!!

Expected Learning Outcomes (HOTS)

- Critically analyze research papers
- Apply ethical theories to technical systems
- Evaluate real-world consequences
- Argue with evidence
- Identify risks
- Recommend governance strategies

Ethical and Explainable AI

Why Ethics in AI? (Need)

- AI influences real human decisions
- Used in hiring, lending, healthcare, governance
- Errors affect lives and rights
- Technical success $\neq$ ethical success

Motivation

- Who is responsible when AI fails?
- Developer, organization, user, government?
- Ethics defines accountability beyond code

Real-World Motivation

- Hiring AI rejecting candidates
- Criminal justice risk scores
- Recommendation systems amplifying harm
- Hidden logic and blind trust

What is AI Ethics?

- Principles guiding responsible AI
- Fair, transparent, accountable, safe
- Applies to design, deployment and governance

Core Ethical Parameters

- Fairness
- Accountability
- Transparency
- Privacy & Consent
- Safety & Reliability
- Explainability
- Human Responsibility

Principle	Short Explanation
Fairness	AI systems should not discriminate against individuals or groups and must provide equal treatment.
Accountability	Humans or organizations must be responsible for AI decisions and outcomes.
Transparency	AI processes and decision-making should be open and understandable.
Privacy & Consent	Personal data must be collected, stored, and used with user consent and protection.
Safety & Reliability	AI systems should operate correctly, securely, and consistently without causing harm.
Explainability	AI decisions should be explainable in a way humans can understand.
Human Responsibility	Humans must oversee AI systems and make final decisions, especially in critical cases.

Fairness

- Avoid discrimination
- Bias from historical data
- Proxy variables cause harm
- Question: who benefits?

Hiring AI ensures equal selection chances for men and women.

Accountability & Responsibility

- AI must be auditable and contestable
- Human-in-the-loop is essential
- Appeals and redress mechanisms required

Bank takes responsibility for wrong loan rejection by AI.

Transparency

- Users should know AI is used
- Understand purpose and logic
- Transparency $\neq$ revealing source code

User is informed that credit approval was decided by an AI system

Privacy & Consent

- Lawful data collection
- Explicit consent
- Data minimization
- Accuracy vs privacy trade-off

Health app asks consent before collecting medical data.

Safety & Reliability

- Robust to errors and misuse
- Critical in healthcare and transport
- Must handle edge cases

Self-driving car safely stops when sensors detect an obstacle.

Explainable AI (XAI)

- AI decisions understandable by humans
- Explains why and how decisions occur
- Essential for trust

Loan AI explains rejection due to low credit score.

Why Explainability Matters

- Detect bias
- Enable accountability
- Allow contestability
- Improve trust

Type of Bias	Description	Example
1 Data Bias	Bias present in the training dataset due to imbalance or missing data	Facial recognition trained mostly on light-skinned faces performs poorly on dark-skinned faces
2 Sampling Bias	Training data does not represent the real-world population	Medical AI trained only on urban hospital data fails for rural patients
3 Algorithmic Bias	Bias introduced by model design, features, or optimization goals	Loan approval system favors high-income groups to maximize profit
4 Historical Bias	AI learns and reinforces existing societal inequalities	Predictive policing targets the same communities repeatedly
5 Labeling Bias	Bias due to subjective or inconsistent human annotations	Emotion detection labels vary based on annotator culture
6 Measurement Bias	Bias caused by incorrect or incomplete measurement methods	Using arrest records as a proxy for crime rate
7 Confirmation Bias	Developers favor outcomes that confirm their assumptions	Ignoring model errors that contradict expected results

8	Interaction Bias	Bias developed through continuous user interaction	Chatbots learn offensive language from users
9	Deployment Bias	Bias occurs when AI is used in a different context than trained	Speech recognition trained on US accents performs poorly on Indian accents
10	Evaluation Bias	Bias due to unfair or incomplete testing benchmarks	Model tested only on majority demographic groups
11	Representation Bias	Certain groups are under-represented in the data	Voice assistants struggle with children's voices
12	Cultural Bias	AI reflects dominant cultural norms	Sentiment analysis misclassifies local slang as negative
13	Temporal Bias	Data becomes outdated over time	Job recommendation AI using pre-pandemic employment data
14	Proxy Bias	Indirect features act as substitutes for sensitive attributes	ZIP code used as proxy for race in loan approvals
15	Value Bias	Bias from designers' values and objectives	Surveillance AI prioritizes security over privacy

Common XAI Approaches

- Feature importance
- Rule-based explanations
- LIME- Local Interpretable Model-agnostic Explanations, SHAP - (SHapley Additive exPlanations)
- Counterfactual explanations

Ethics + Explainability

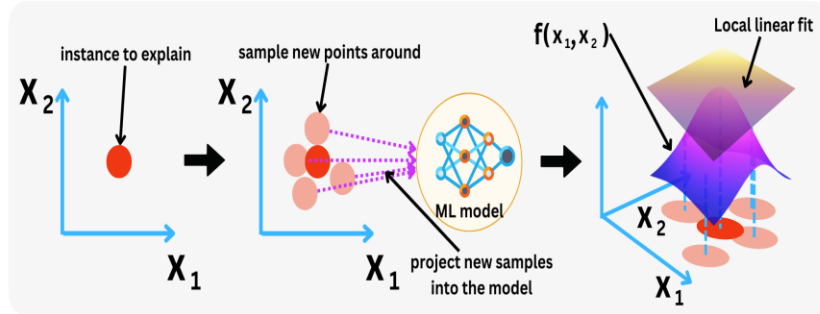
- Ethics defines what should be done
- Explainability shows why it was done
- Both required for trustworthy AI

Indian Context – AI Ethics

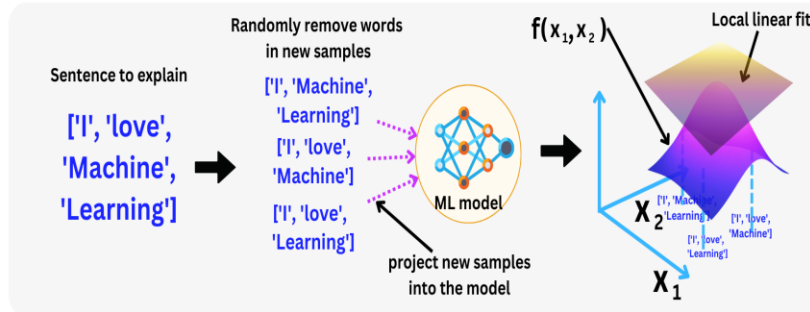
- Large-scale public deployment
- High impact on citizens
- Explainability critical for inclusion

Explainable AI: LIME

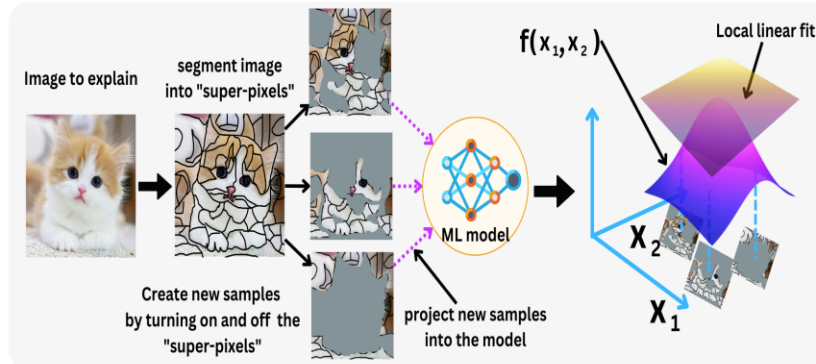
LIME with Tabular data



LIME with Text data



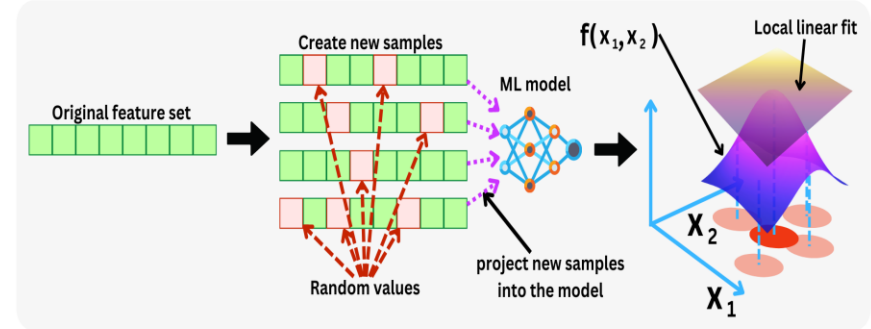
LIME with Image data



Explainable AI: SHAP

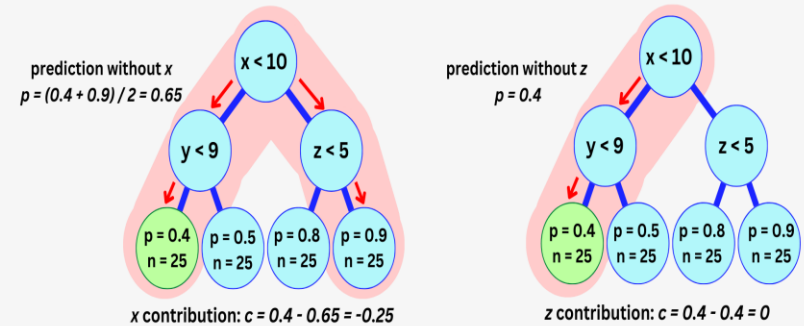
TheAiEdge.io

Kernel SHAP: LIME with Shapley Smoothing Kernel



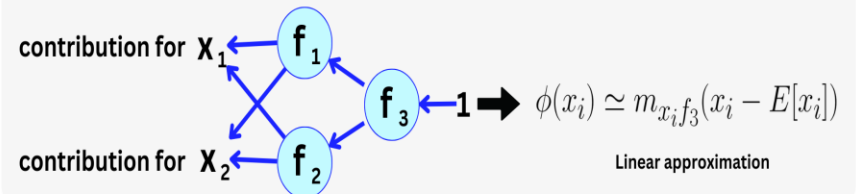
Tree SHAP: Shapley estimates for Trees

instance: $[x = 8, y = 5, z = 12]$ prediction with all the features: $p = 0.4$



Deep SHAP: Shapley estimates for Neural Networks

Backpropagate Shapley contributions



India-Specific Frameworks

- NITI Aayog – Responsible AI
- MeitY – IndiaAI Mission
- DPDP (Digital Personal Data Protection) Act 2023

Role of Future Engineers

- Question data and assumptions
- Avoid blind trust
- Design human-centric AI
- Recommend governance

Questions?



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